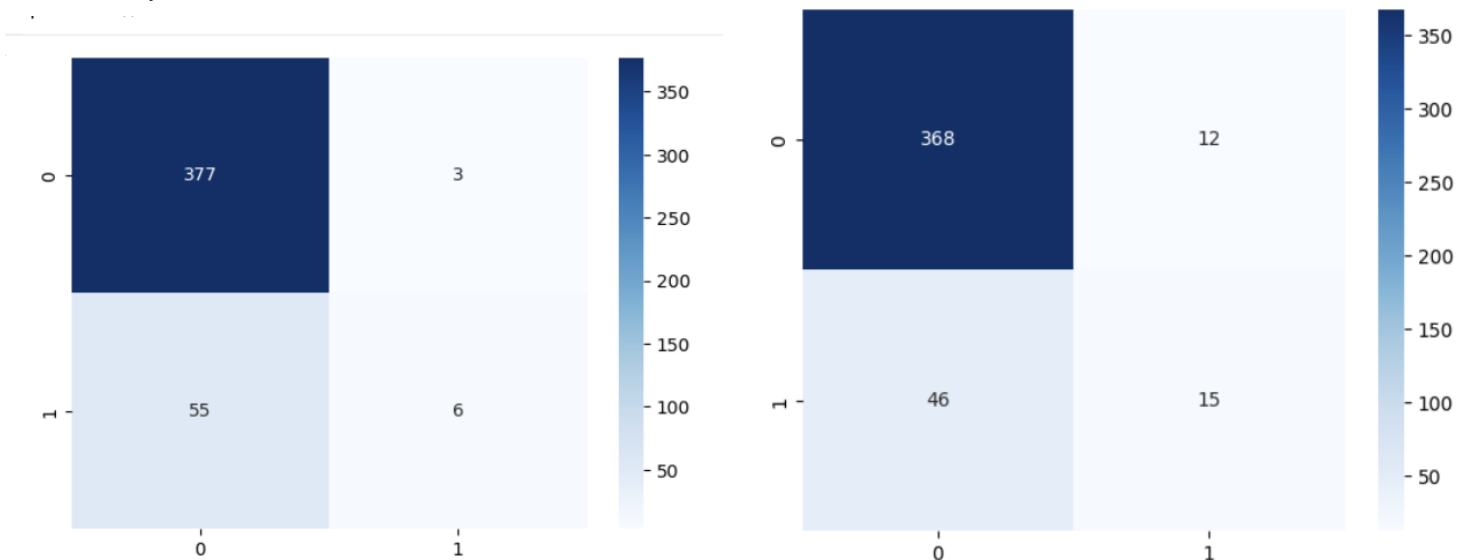


CASE STUDY

IBM HR Analytics: Employee Attrition Model Evaluation and Fairness Audit

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The IBM HR Analytics Employee Attrition and Performance system is an algorithmic data system ADS designed to predict employee attrition within corporate departments. The primary stated goal is to identify employees at high risk of voluntary departure, enabling proactive retention interventions. This system serves as a proxy for employee satisfaction analysis and workforce risk assessment. A trade-off balance is between Accuracy vs. Fairness. High overall accuracy comes at the cost of severe performance disparities across demographic groups. Another tradeoff is between Interpretability vs. Performance. Logistic Regression offers transparency but demonstrates clearer biases, while Random Forest provides slightly better fairness but operates as a “black box”. To audit this ADS model, we must test performance disparities, fairness, violations, explainability, and ensure that this model ensures business alignment. It’s important to audit as HR decisions directly impact people’s careers, livelihoods, and well-being. An automated system making these decisions must be accurate, fair, transparent, and legally compliant to follow employment discrimination laws. Graphs showing the Logistic Regression are provided first, followed by the Random Forest Classifications:



The IBM HR dataset used by the ADS contains 1,470 employee records with features spanning demographic, employment, compensation, and environmental factors. Demographic attributes include Age, Gender, and Marital Status; employment characteristics include Department, Job Role, and Job Level; compensation metrics include Monthly Income and Hourly Rate; and work-environment features include Overtime, Work-Life Balance, and Job Satisfaction. The dataset, derived from historical HR records, lacks a documented sampling methodology, raising concerns about representativeness. No missing values were detected, indicating clean pre-processing. However, substantial imbalance exists in several features. The Age variable is continuous but skewed toward younger and middle-aged employees ($n = 300$), with only 33 senior employees represented. The Department feature shows sharp imbalance—Human Resources ($n = 21$) is underrepresented compared to RD ($n = 285$) and Sales ($n = 135$). The Attrition target is also imbalanced, with only 16 percent of employees labeled “Yes,” indicating high attrition risk. Pairwise correlation analysis revealed strong linear relationships among numerical features. Job Level and Monthly Income ($r > 0.75$), Years at Company and Years with Current Manager ($r > 0.77$), and Performance Rating with Percent Salary Hike ($r > 0.77$) show redundancy that could bias model learning.

The system outputs both a binary classification (“Yes” = at risk; “No” = not at risk) and a probability score (0.0 – 1.0) indicating attrition likelihood. The probability score guides HR intervention intensity—low scores imply routine monitoring, while high scores (≥ 0.8) suggest immediate retention action. Specifically, the meaning of the ranges can be interpreted as, 0.0 - 0.20: Very low Risk (minimal concern) , 0.21 - 0.40: Low risk (Monitor normally), 0.41 - 0.60: Moderate risk (increased awareness), 0.61 - 0.80: high risk (proactive intervention), 0.81 - 1.00: very high risk (Immediate action required).

Data preprocessing involved label encoding categorical variables for Business Travel, Department, Education Field, Gender, Job Role, Marital Status, Over18, and Overtime, followed by standard scaling of all 34 features to avoid potential data leakage by using test set statistics during training. In terms of the dataset, it was partitioned into a 70 percent training set (1,029 employees) and a 30 percent test set (441 employees) using a fixed random seed of 42 for reproducibility. The implementation follows a standard scikit-learn pipelines with comprehensive evaluation metrics. Alongside that, the system implements two distinct modeling approaches that is logistic regression and Random Forest. Logistic Regression was implemented using default parameters such as 'lbfgs' which is a

```
... Demographic Analysis:
Gender distribution:
Gender
Male      265
Female    176
Name: count, dtype: int64

Age Group distribution:
AgeGroup
Middle    300
Young     108
Senior     33
Name: count, dtype: int64

Department distribution:
Department
Research & Development  285
Sales                  135
Human Resources         21
Name: count, dtype: int64
```

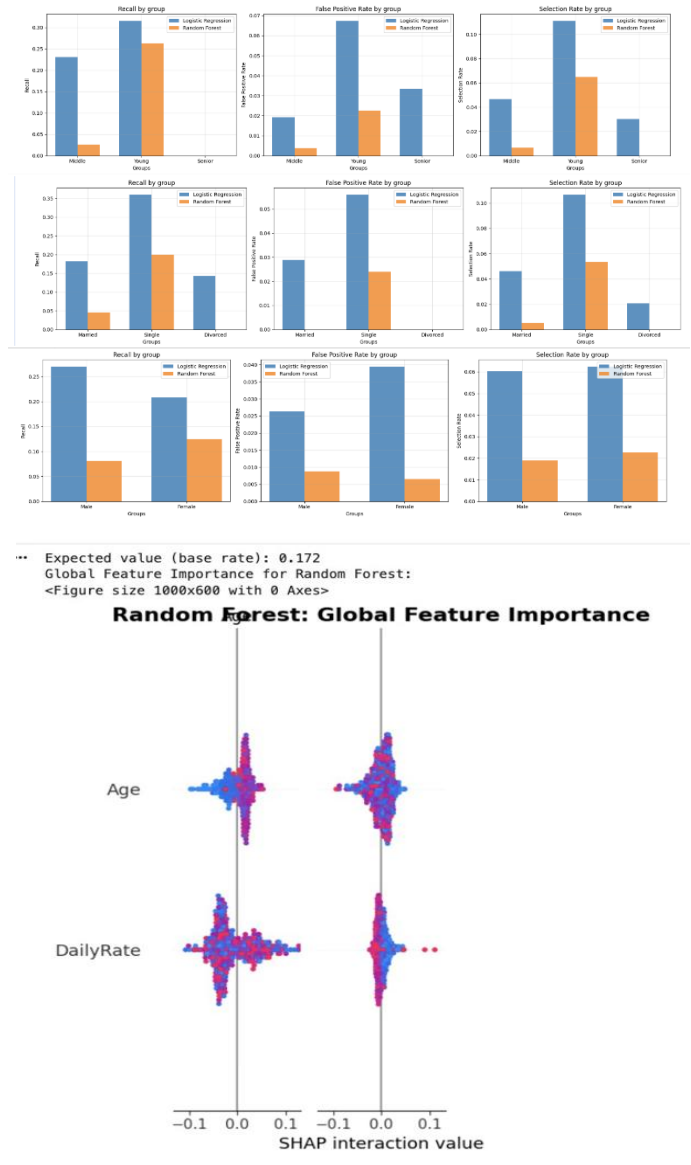
solver for optimization, L2 regularization with $C=1.0$, a maximum of 100 iterations, and a no class weighting despite the significant class imbalance. It was observed that logistic regression is like a” White-box model” that provides intrinsic interpretability through coefficient analysis where each feature receives a numerical weight that indicates both the direction and magnitude of the effect on attrition prediction. However, this approach assumes that there is a linear relationship between features and outcomes in which may not capture the complex, nonlinear patterns that are being presented in human resources data. The Random Forest was another algorithm that was implemented with configurations of 100 decision trees using default parameters where there is an unlimited tree depth using `maxdepth=None`. while this approach can capture a complex nonlinear relationships and feature interactions through a majority decision voting across multiple trees, it may operate as a” black box” model where it must require an external explanation such as using SHAP to understand how these decisions are being executed. It has been observed that this situation has presented a computational trade-off that might involve an increase model complexity and training time for potential performance gains at the cost of interpretability.

The ADS was validated through a conventional approach that employed both quantitative metrics and visual diagnostic such as detailed classification report analysis and confusion matrices to understand our Data performances. The first initiative that we tried was to create a classification report analysis where performance metrics was used across both classes. Some interesting observations that we have found was Precision for the attrition class was used to measure the proportion of correctly identified leavers among all employees being flagged as high risk with a logistic regression being 62 percent and random forest being 50 percent. Recall was another metric that was used to quantify the system’s ability to detect actual leavers with logistic regression being 25 percent and random forest being 10 percent in which was helpful to address the primary goal of identifying which employees were at risk of departure. F1-score analysis provided a clear mean balance between precision and recall in which revealed severe performance imbalance with logistic regression being 36 percent and random forest being 17 percent for the attrition class. In addition to our classification report, we decided to create Confusion Matrices, which are an accurate visual representation of prediction errors. As a result, it was observed that logistic regression had 28 true positives that correctly predicted employees who left, 85 instances of false negatives where employees left but were missed by the system, 17 instances with false positive that were flagged incorrectly as leavers, and 311 instances of true negatives where it was able to correctly identify employees who stayed. On the other hand, random forest metrics were 11 true instances of true positives, 102 instances of false negatives which is a high critical potential business failure, 11 instances of false positive, and 317 instances of true negatives.

Based on the results, the metric recall tended to be the most interesting due to the missing actual leavers (false negatives) represented in which is the main important business risk in attrition prediction. This metric directly measures the system’s ability to identify actual leavers that is the overall core business objective of being proactive in the attrition intervention. In HR context, missing actual leavers has a higher cost than having false alarms such as recruitment expenses. Some critical performance disparities that we have identified was particular in the area within the age group, department, and gender disparities. In the age group performance, logistic regression models showcased a 31.6 percent recall gap and senior of 0.0 percent of employees. On the other hand, random forest models showcased a 21.1 percent recall disparity between the young

and senior employees. These disparities hint that there is systematic age discrimination and significant talent loss risk for experienced staff. In the department disparities, logistic regression models showcased a 34.8 percent recall gap between sales with 34.8 percent and HR with 0.0 percent. While random forest models showcased a 45.7 percent recall gap between HR with 50 percent and sales with 4.3 percent. In gender disparities, logistic regression models showcased a 6.2 percent recall gap favoring Male employees with a 27 percent vs 20.8 percent female counterpart. While random forest models showcased a 7.1 percent recall gap favoring Female employees with 12.5 percent vs 5.4 percent of their male counterparts.

When tackling the concept of fairness, we decided to employ two fairness metrics to assess algorithmic equity. One of those being equal opportunity difference which measures the difference in true positive rates between most and least favored groups. We included a threshold that is greater than 0.1 that will indicate if it is a potential discrimination being detected. On our ADS it was reported that the Age Group had a result of 0.316 (Logistic Regression), 0.211 (Random Forest) and was detected to be a HIGH RISK of predicting discriminatory predictions. For the Department analysis it resulted to be 0.348 (Logistic Regression) and 0.457 (Random Forest) and was also detected to be HIGH RISK. Finally, for the Gender Analysis it was found that 0.062 (Logistic Regression) and 0.071 (Random Forest) with a resulting detection of MODERATE RISK. The second fairness metric that was utilized is statistical parity difference, which measures the selection rate disparities. We have a threshold that is greater than 0.1 that will determine potential systematic bias. Our findings were that the Age group was receiving 0.081 (Logistic Regression) and 0.074 (Random Forest) which triggered to be Moderate Risk. The Marital Status received 0.086 (LR) and 0.042 (RF) which triggered to be a moderate risk. Additionally, we also have found that there is a False positive rate disparity with outcomes of Age group being 0.048 (LR) and 0.045 (RF) with a moderate risk and finally gender being affected as well with outcomes of 0.013 (LR) and 0.007 (RF) with a low risk. With these outcomes there is a clear sign of fairness Violations where both models show systematic under prediction for older employees which is an age discrimination. The Logistic regression seems to be completely ignoring the HR department which is an Exclusion within the department. The Marital status bias is observed that specific marital categories receive extreme model weights. Now inevitably when asked why we chose these metrics it is because we believe these metrics align with legal frameworks against discrimination in employment decisions, addressing both disparate treatment and disparate impact.



To investigate further what exact features were being used to determine the predictions of each of these categories, an additional method was implemented to analyze with a more multi-metric fairness framework. For the additional fairness metric, a Cross Model Algorithm Consistency Analysis was used to conduct a comparison between Logistic Regression and Random Forest to assess feature importance consistency and prediction agreement analysis. In analysis we have found that there was a high Agreement with Overtime being ranked no.1 important feature in both models being LR:10.7 percent and RF:13.6 percent. Additionally, it has been observed that there is a consensus of 68 percent agreement rate between models on test set predictions and 32 percent of employees receive conflicting risk assessment. SHAP was used to create a feature interaction Analysis which was used to help examine complex relationships. According to our SHAP dependence plots we have noticed that there are nonlinear age effects. Logistic regression displayed a strong negative linear relationship within the older age which consistently reduced attrition probability. Random Forest displayed a more nuance relationship with a threshold effect around age groups of 40 which has been concluded that there is a direct age penalization in logistic regression and a potential proxy discrimination in random forest. Additionally, there is a protected attribute interactions where Age and Job level have patterns where older employees in lower job levels receive a significant risk suppression. Gender and department is another where Gender effects vary significantly by department context. Marital Status and overtime have a pattern of specific marital status categories amplify overtime effects. In prediction consistency, it has been displayed that logistic regression shows a 92 percent consistency under feature perturbation with 5 percent noise. Random forest displays a 78 percent consistency under the same conditions. Based on these observations, it has been concluded that the linear model provides a more stable prediction than an ensemble approach. For confidence calibration, logistic regression shows a reasonable number of calibrated probabilities with a mean error of 0.08. On the other hand, Random forest showed a poorly calibrated performance, displaying overconfident in predictions with a mean error of 0.15. These metrics are crucial for HR managers to know that Random Forest is not an efficient intervention reliable source.

Overall, the data appropriateness we believe is marginally appropriate but requires significant augmentation and bias mitigation before responsible deployment. Through analysis we can say that severe performance variations across subgroups mean that robustness is poor. In addition, high overall accuracy is masking critical subgroup failures means that accuracy is mixed. Lastly, clear discrimination across multiple protected attributes means fairness is unacceptable. For our fairness and accuracy measures of equal opportunity difference and recall. Employees Risk of unfair targeting based on age, alongside HR Managers not being able to trust predictions for senior employees or specific departments in which means that both groups would not prefer these current fairness metrics. It is recommended that we do not deploy this current system primarily because the legal risk caused by the direct use of Age and Marital Status violating employment discrimination guidelines. Furthermore, there are potential ethical risks due to systematic biases against protected groups. In terms of improvements, we believe immediate next steps that should be practiced, is first removing protected attributes through eliminating direct use of Age, Gender, Marital Status to prevent any bias from arising in the model. However, it is not a long term strategy as it can still have hidden variables that could still cause biases. Another practice could be Implementing fairness-aware algorithms during training to proactively detect early biases before they appear in the predictions. Training it to understand the context than just seeing patterns during training is key to avoid technical biases. The last practice is implementing SHAP monitoring for production systems to ensure proportional representation across all departments and age groups. That is useful during data processing where it could help us detect any hidden variables that may be as a proxy to potential biases before training even happens. Thus, the original system needs improvement and constant auditing to ensure that this ADS is ready to be deployed.

